SLOW CENTURIES · STUDY GUIDE

Watch the world's systems take shape.

Timelapse of Algorithms

Four thousand years of step-by-step thinking, from clay to code.



EPISODE LENGTH
about 12 minutes

SUGGESTED LEVEL
grades 7 to 12, intro college

SUBJECTS
math, history, computer science

QR code appears after setup
WATCH

QR code appears after setup
ALL LINKS

1800 BCE — **TODAY**

1 In one paragraph

An algorithm is a finite list of steps that solves a problem. This episode follows that idea for about 4,000 years. It starts with a Babylonian clay tablet that gives the square root of two and Egyptian scribes who multiplied by doubling. It moves through Euclid, Archimedes, Liu Hui, Aryabhata and al-Khwarizmi, whose name became the word "algorithm." Then the steps move into machines, from Jacquard's punched-card loom and Ada Lovelace's Note G to Turing, ENIAC and Grace Hopper's compiler. It ends with the algorithms that run today's world, from shortest routes and sorting to search and machine learning.

2 Watch with chapters

0:00 **Before there were machines**
What makes a set of steps an algorithm

0:41 **1. Clay and Papyrus**
The square root of two in clay, and multiplying by doubling

2:17 **2. Proof and Approximation**
Euclid, Archimedes, Liu Hui and Aryabhata

4:08 **3. The Name**
al-Khwarizmi, Fibonacci and the Maya Venus table

6:00 **4. Pattern and Thread**
Chokwe *sona*, binary, the Jacquard loom and Ada Lovelace

7:43 **5. Logic and Machines**
Boole, Hollerith, Turing, ENIAC and Grace Hopper

9:38 **6. Search and Learning**
Dijkstra, Quicksort, P versus NP, the Perceptron, PageRank and AlexNet

11:17 **A line in the sand**
What has and hasn't changed

3 The timeline

c. 1800–1600 BCE — Mesopotamia
A school tablet (YBC 7289) gives the square root of two to about six decimal places, written in base sixty.

c. 1550 BCE — Egypt
The scribe Ahmes copies the Rhind Mathematical Papyrus, a workbook of problems and the steps to solve them.

c. 300 BCE — Alexandria
Euclid's *Elements*, Book VII, describes how to find the greatest common divisor of two numbers.

c. 250 BCE — Syracuse
Archimedes traps pi between 3 10/71 and 3 1/7 using polygons with up to 96 sides.

263 CE — China
Liu Hui's commentary on the *Nine Chapters* pushes the polygon method past 3,000 sides and gives pi as 3.1416.

499 CE — India
Aryabhata's *Aryabhatiya* includes the method later called *kuttaka*, "the pulverizer."

c. 825 CE — Baghdad
Muhammad ibn Musa al-Khwarizmi writes on calculating with the Indian numerals, including zero.

1100s — Europe
A Latin translation opens "*Dixit Algorizmi*," "so says al-Khwarizmi." His name becomes *algorism*, then *algorithm*.

1202 — Pisa
Leonardo of Pisa (Fibonacci) teaches the new numerals to European traders in *Liber Abaci*.

c. 1200s — Maya world
The Dresden Codex Venus table tracks Venus and builds in corrections to stay accurate.

Long-standing — Northeastern Angola
Chokwe storytellers draw *sona*, rule-based line drawings in the sand.

1679–1703 — Germany
Gottfried Wilhelm Leibniz writes every number with only 1 and 0.

1804 — Lyon
Joseph Marie Jacquard's loom is controlled by a chain of punched cards.

1843 — London
Ada Lovelace's Note G sets out how the Analytical Engine could calculate Bernoulli numbers.

1854 — Cork
George Boole shows that logic can be written as algebra.

1890 — United States
Herman Hollerith's punched-card machines tabulate the census.

1936 — Cambridge
Alan Turing defines what it means for a problem to be computable.

1941 — Ottawa
Canada opens the Examination Unit to break enemy codes.

1945 — Philadelphia
Six women program ENIAC by setting switches and plugging cables.

1952 — United States
Grace Hopper builds one of the first compilers.

1956 — Amsterdam
Edsger Dijkstra works out a shortest-path method.

1958 — Cornell
Frank Rosenblatt builds the Perceptron, a machine that learns from examples.

1959 — Moscow
Tony Hoare invents Quicksort.

1971 — Toronto
Stephen Cook poses the question now called P versus NP.

1998 — Stanford
Larry Page and Sergey Brin describe PageRank.

2012 — Toronto
AlexNet, by Krizhevsky, Sutskever and Hinton, starts the deep learning era.

2024 — Stockholm
John Hopfield and Geoffrey Hinton share the Nobel Prize in Physics.

4 Key terms

Algorithm. A finite list of clear steps that solves a problem. Anyone who follows the steps gets the same result.

Loop. A step, or group of steps, that repeats until a condition is met. The Babylonian square-root method is a loop: guess, divide, average and repeat.

Base sixty. A number system that counts in groups of sixty. We still use it for minutes, seconds and the degrees in a circle.

Greatest common divisor. The largest number that divides two other numbers with nothing left over. Euclid's method finds it.

Approximation. An answer that gets closer and closer to the true value without reaching it exactly, like Archimedes' bounds on pi.

Place value. A system where a digit's position sets its value. In 305, the 3 means three hundreds because of where it sits.

Binary. A number system that uses only two digits, 1 and 0. Modern computers store everything this way.

Compiler. A program that turns instructions written for people into code a machine can run.

Computable. A problem is computable if some algorithm can always solve it. Turing proved that some problems are not.

Neural network. A program made of many simple connected units that adjusts its own settings by learning from examples.

5 Hands-on activity

Multiply like an Egyptian scribe

TIME: 15 MINUTES · MATERIALS: PAPER AND PENCIL

1. Pick two numbers, for example 12 × 13.
2. Make two columns. Start the left column at 1 and the right column at the first number (12).
3. Double both columns on each new row: 1 and 12, 2 and 24, 4 and 48, 8 and 96. Stop before the left column passes the second number (13).
4. Find the left-column numbers that add up to the second number: 1 + 4 + 8 = 13. Tick those rows.
5. Add the right-column numbers on the ticked rows: 12 + 48 + 96 = 156. That's the answer.

WORKED EXAMPLE · 12 × 13

	COUNT	DOUBLE
✓	**1**	**12**
	2	24
✓	**4**	**48**
✓	**8**	**96**

1 + 4 + 8 = 13 → 12 + 48 + 96 = 156

YOUR TURN · ____ × ____

	COUNT	DOUBLE
	1	
	2	
	4	
	8	
	16	

Follow-up questions

- Why can every whole number be built from 1, 2, 4, 8, 16 and so on, each used at most once?
- Write 13 in binary (1101). How do the ticked rows match the 1s?
- Is this method an algorithm? Point to the steps, the loop and the stopping rule.

Extension: try Euclid's method on 84 and 36. Take the smaller from the larger and repeat with what's left until the two numbers match. You should reach 12.

6 Discussion questions

1. The episode says an algorithm is "a finite list of steps that solves a problem." Which examples in the video fit that definition best? Do any stretch it?
2. Why do you think written calculation on paper slowly beat the abacus in Europe, even though the abacus was fast?
3. Ada Lovelace is "often called" the first programmer. Why might historians choose those words carefully?
4. The ENIAC programmers were left out of the story for decades. What does it change when we learn their names?
5. Turing proved that some problems can't be solved by any algorithm. Why might that be a useful thing to know?

7 What's still debated

- **How the Babylonian scribe found √2.** We have the answer on the tablet but not the working. The "guess, divide, average" loop is one method often linked to Babylon. It isn't proven to be the one used.
- **Who counts as "the first programmer."** Lovelace's Note G is a clear step-by-step program for a machine. Some historians point out that Babbage had written earlier examples for the same engine.
- **P versus NP.** Nobody knows yet whether every problem that's easy to check is also easy to solve. It's one of the biggest open questions in mathematics.

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8 Primary sources and places to visit online

Each name below is a link to the institution that holds the object or published the original work. Extra links follow after "Also". Links were checked on 30 October 2026. If one breaks, search the name on the institution's site.

1. Clay and Papyrus

YBC 7289, the square-root-of-two tablet · `isaw.nyu.edu`, Yale Babylonian Collection. Also: catalogue record and photos (CDLI) · `cdli.earth`

The Rhind Mathematical Papyrus · `britishmuseum.org`, British Museum, London.

2. Proof and Approximation

A papyrus fragment of Euclid's *Elements* · `penn.museum` (Papyrus Oxyrhynchus 29, about 100 CE), Penn Museum, Philadelphia.

The Archimedes Palimpsest · `archimedespalimpsest.org`, a medieval copy of Archimedes' works, including texts found nowhere else, recovered with special imaging.

Aryabhata's *Aryabhatiya*. An English translation by W. E. Clark (1930) is in the public domain. Search Wikisource or the Internet Archive for "The Aryabhatiya of Aryabhata."

3. The Name

The "*Dixit Algorizmi*" manuscript · `cudl.lib.cam.ac.uk`, the Latin al-Khwarizmi text, Cambridge University Library, MS Ii.6.5.

The Dresden Codex · `loc.gov`, including the Venus table. Full digital copy via the Library of Congress. The original is at the Saxon State and University Library, Dresden.

4. Pattern and Thread

Chokwe *sona*. Paulus Gerdes, *Geometry from Africa: Mathematical and Educational Explorations* (Mathematical Association of America, 1999). Please read the note on cultural respect below before teaching this section.

Leibniz, "Explication de l'Arithmétique Binaire" (1703) · `journals.openedition.org`, annotated edition with facsimile, BibNum (in French).

The Jacquard hand loom · `collection.sciencemuseumgroup.org.uk`, Science Museum Group, UK. Also: Jacquard punched cards · `collection.sciencemuseumgroup.org.uk`

Babbage's Analytical Engine, trial model · `collection.sciencemuseumgroup.org.uk`, Science Museum Group.

Ada Lovelace's papers online · `claymath.org`, Clay Mathematics Institute. Also: Bodleian Libraries blog on Lovelace's papers · `blogs.bodleian.ox.ac.uk`

5. Logic and Machines

George Boole, *An Investigation of the Laws of Thought* (1854) · gutenberg.org, free full text, Project Gutenberg.

Hollerith's tabulating machine · si.edu, Smithsonian National Museum of American History.

Alan Turing's 1936 machine · turing.org.uk, The Alan Turing Internet Scrapbook, by biographer Andrew Hodges.

The Examination Unit (1941–1945) · canada.ca, Parks Canada. Also: Communications Security Establishment: Our story · cse-cst.gc.ca

An ENIAC accumulator panel · si.edu, Smithsonian National Museum of American History.

Grace Hopper completes the A-0 compiler · computinghistory.org.uk, Centre for Computing History.

6. Search and Learning

Dijkstra (1959), "A note on two problems in connexion with graphs" · link.springer.com, *Numerische Mathematik*.

Stephen Cook and "The Complexity of Theorem-Proving Procedures" (1971) · heidelberg-laureate-forum.org, Heidelberg Laureate Forum profile.

Rosenblatt's Perceptron · news.cornell.edu, Cornell Chronicle: "Professor's perceptron paved the way for AI – 60 years too soon."

Brin and Page (1998), the paper that introduced Google and PageRank · stanford.io, Stanford.

Krizhevsky, Sutskever and Hinton (2012), the AlexNet paper · papers.nips.cc, NeurIPS proceedings.

The 2024 Nobel Prize in Physics · nobelprize.org, Nobel Prize press release.

The whole episode

Revolution: The First 2000 Years of Computing · computerhistory.org, an online exhibit from the Computer History Museum.

9 A note on cultural respect

The Chokwe sona tradition and the Maya Dresden Codex belong to living cultures. Present them as knowledge systems in their own right, not as early versions of European mathematics. Where you can, use sources written by or with members of those communities.

10 Further reading

- Jean-Luc Chabert (ed.), *A History of Algorithms: From the Pebble to the Microchip* (Springer, 1999).
- Eleanor Robson, *Mathematics in Ancient Iraq: A Social History* (Princeton University Press, 2008).
- Kathy Kleiman, *Proving Ground: The Untold Story of the Six Women Who Programmed the World's First Modern Computer* (Grand Central, 2022).

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